Extension: Contextual Thompson Sampling

William Dennis

June 2026

The MVP agents are state-blind bandits. Contextual Thompson Sampling conditions action selection on the current activity level, learning per-context intervention effectiveness.

1 Formulation

Context $c_t = s_t.\texttt{activity} \in \{\texttt{sedentary}, \texttt{active}\}$. Posterior pairs indexed by (c, a):

$$(\alpha_{c,a},\ \beta_{c,a}) \quad \forall\, c \in \mathcal{C},\ a \in \mathcal{A}$$

Action selection samples $\theta_{c_t,a} \sim \text{Beta}(\alpha_{c_t,a}, \beta_{c_t,a})$ and picks $a^* = \arg\max_a \theta_{c_t,a}$. Update applies only to the observed (c_t, a_t) pair.

2 Theoretical Optimal

Under the rule-based transition probabilities:

$$P_{\text{nudge}} = \begin{bmatrix} 0.7 & 0.3 \\ 0.5 & 0.5 \end{bmatrix} \qquad P_{\text{idle}} = \begin{bmatrix} 0.9 & 0.1 \\ 0.4 & 0.6 \end{bmatrix}$$

the optimal contextual policy is: nudge when sedentary ($0.3 > 0.1$), idle when active ($0.6 > 0.5$). The stationary distribution under this policy satisfies $\pi_s = 4/7$, $\pi_a = 3/7$, giving:

$$E[r] = \tfrac{4}{7} \cdot 0.3 + \tfrac{3}{7} \cdot 0.6 = \tfrac{3}{7} \approx 0.429 \text{ per step} \implies 192.9 \text{ total}$$

For comparison, the best non-contextual policy (always nudge) achieves $E[r] = 0.375$ per step (168.8 total). The contextual upper bound is 14.3% higher.

3 Results

10 seeds, 450 timesteps each.

The contextual agent learns that nudging sedentary users is effective while nudging active users wastes intervention budget. Standard TS cannot represent this distinction.

4 Implementation

Extended `ThompsonSamplingAgent` with `contextual` and `context_feature` parameters. When `contextual=False` (default), behaviour is identical to standard TS. Config:

Agent	Total Reward	Per Step	vs Optimal
Contextual TS	170.8 ± 15.6	0.380	-11.4%
Standard TS	157.8 ± 14.8	0.351	—
Contextual optimal (theory)	192.9	0.429	—
Non-contextual optimal	168.8	0.375	—

Table 1: Contextual TS beats standard TS by 8.2%. The 11.4% gap to the contextual upper bound reflects incomplete convergence in 450 steps across 4 posterior pairs.

```
agents:
  - type: thompson_sampling
    alpha_prior: 1
    beta_prior: 1
    contextual: true
    context_feature: activity
```

5 Design Decisions

Question	Decision	Status
Which context feature?	`activity` – already in StateView, no state changes needed	Confirmed
Why not `step_of_day`?	5 values $\times$ 2 actions = 10 posteriors. Deferred.	Deferred
ABC vector in/out?	Not yet. Discrete context works with current string-in/string-out ABC.	Deferred